Waves at boundaries

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe reflection and transmission at a boundary.

SUCCESS CRITERIA - I can...

1. Identify incident, reflected and transmitted waves.
2. Distinguish fixed-end from free-end pulse reflection.
3. Track energy without assuming equal amplitudes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Echoes return at 0.003 s and 0.005 s with the same wave speed. Compare ranges. Does delay alone determine mass?

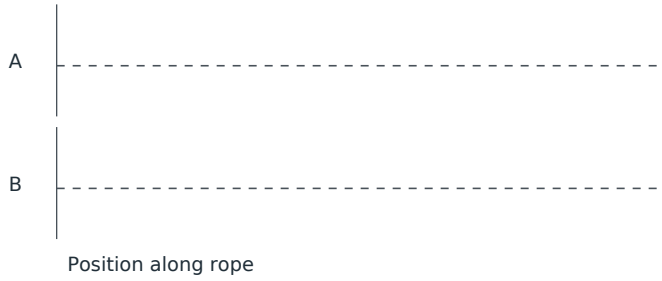
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $f_{reflected} = f_{incident}$.
At a stationary boundary. A displacement pulse returns inverted from an ideal fixed end and upright from an ideal free end.

Worked reasoning: At a fixed end, an upward rope pulse reflects downward. At an ideal free end it reflects without inversion. These are different boundary conditions.

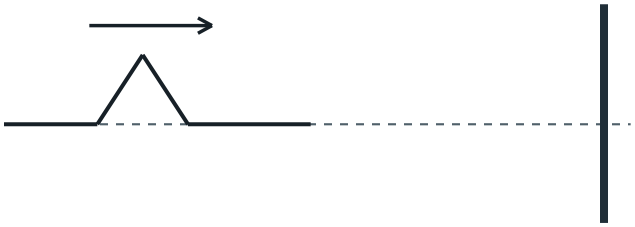
Your turn: Draw incident and reflected pulse snapshots for both ends; show propagation directions.



Marzano check - circle: ① ② ③

03 Read and interpret the model

An upward rope pulse reaches an ideal fixed end. Sketch the reflected pulse and its travel direction.



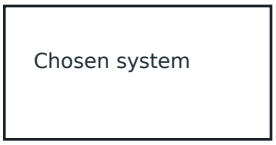
Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A pulse carries 0.80 J to a boundary. Of this, 0.20 J is reflected and 0.10 J is absorbed. Complete the energy-flow model. Find transmitted energy and the reflected, absorbed and transmitted fractions.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Does a smaller transmitted amplitude prove frequency changed? Explain what usually fixes the frequency at a stationary boundary.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 0.4 kg vapor releases 120000 J while condensing at constant temperature. Find latent heat and signed Q for the sample.
Label temperatures, samples and relevant energy transfers.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

At a stationary boundary frequency stays 5 Hz while wave speed changes from 10 to 20 m/s. Find wavelength on each side.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A pulse reflects from a fixed end and meets an equal incident pulse. At an instant their displacements are +2 and -2 cm at one point. Find total displacement and explain why zero there does not mean all energy vanished.



Sketch and label your response.

Marzano check - circle: 1 2 3